

QA LAB BRASIL

Reference guide - QA Lab Brasil

VS Code, PowerShell and Terminal

Development environment, terminal, PATH, variables, venv, pip and useful QA commands.

VS Code

Visual Studio Code is an extensible editor widely used for automation. It supports code editing, extensions, debugging, Git and integrated terminals.

PowerShell, CMD and Git Bash

CMD is the traditional Windows shell. PowerShell combines a shell with an automation language. Git Bash provides a Bash-like environment. The VS Code terminal simply hosts one of these shells.

PATH

PATH is an environment variable that tells the system where to look for executables. If python, git or java is not recognized, installation or PATH may be incorrect.

Python virtual environment

`python -m venv .venv` creates an isolated environment. In PowerShell, `.\.venv\Scripts\Activate.ps1` activates it. This prevents dependency conflicts between projects.

pip and requirements.txt

pip installs Python packages. `requirements.txt` records dependencies so another computer can reproduce the environment with `pip install -r requirements.txt`.

Environment variables and secrets

Credentials and tokens should not be hardcoded. Environment variables and CI/CD secrets provide sensitive values at runtime.

Debugging in VS Code

Breakpoints, variable inspection and step-by-step execution help investigate automation. The terminal and debugger solve different problems.

First diagnostics

Use version commands such as `python --version`, `pip --version`, `git --version`, `java -version` and `node --version`. Knowing which executable is used reduces configuration issues.

QA Tip: After installing a tool, validate it in the terminal with its version command before configuring the project.

Official: <https://code.visualstudio.com/>

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